

CI/CD Pipeline for Earthquake API Data Persistence Using AWS Lambda and DynamoDB

Objective

Implement an automated CI/CD pipeline using GitHub Actions to deploy an AWS Lambda function that fetches earthquake data from the USGS API and stores it in Amazon DynamoDB.

Technologies

Python, GitHub, GitHub Actions, AWS Lambda, Amazon DynamoDB, IAM, USGS Earthquake API

Architecture

Developer -> GitHub Push -> GitHub Actions -> Lambda Deployment -> Earthquake API -> DynamoDB

Repository Structure

.github/workflows/deploy.yml; Earthquake_CICD/README.md; lambda_function.py; requirements.txt

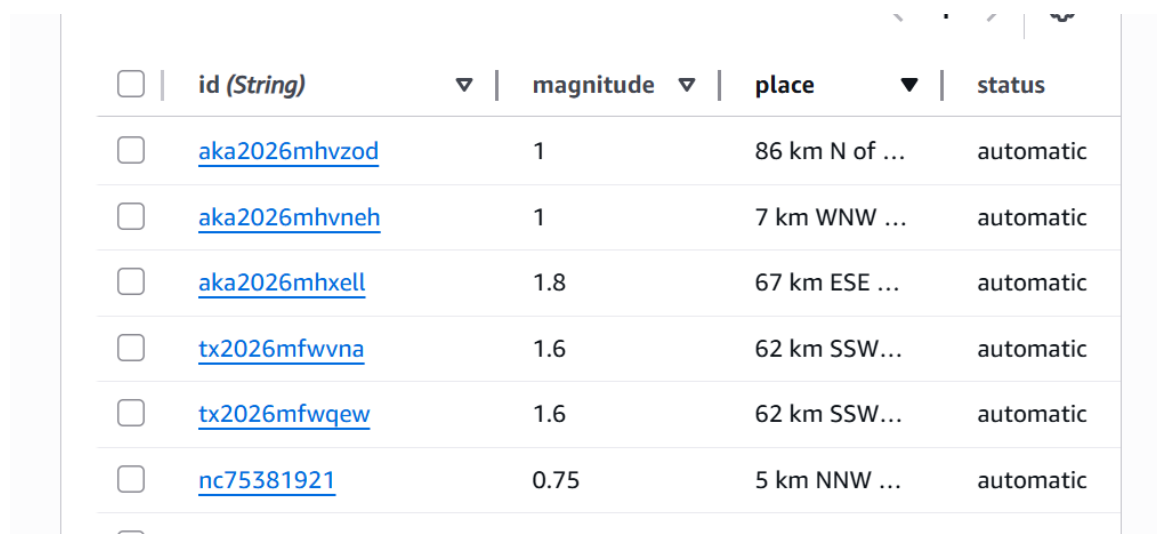
Implementation

Lambda fetches latest earthquake records and stores them in DynamoDB. GitHub Actions deploys on every push.

Result

Lambda test succeeded with statusCode 200 and records_inserted 20. DynamoDB contains earthquake records.

DynamoDB Screenshot



The screenshot shows the AWS DynamoDB console interface. At the top, there are navigation icons for back, forward, and search. Below the navigation bar, there is a table with the following columns: 'id (String)', 'magnitude', 'place', and 'status'. Each column has a dropdown arrow. The table contains seven rows of data, each with a checkbox on the left. The data rows are:

	id (String)	magnitude	place	status
☐	aka2026mhvzod	1	86 km N of ...	automatic
☐	aka2026mhvneh	1	7 km WNW ...	automatic
☐	aka2026mhxell	1.8	67 km ESE ...	automatic
☐	tx2026mfwvna	1.6	62 km SSW...	automatic
☐	tx2026mfwqew	1.6	62 km SSW...	automatic
☐	nc75381921	0.75	5 km NNW ...	automatic